illustrated in Fig. 3-6a, with the corresponding frequency spectrum in (b). As just described, the original analog signal can be perfectly reconstructed by passing this impulse train through a low-pass filter, with the cutoff frequency equal to one-half of the sampling rate. In other words, the original signal and the impulse train have identical frequency spectra below the Nyquist frequency (one-half the sampling rate). At higher frequencies, the impulse train contains a duplication of this information, while the original analog signal contains nothing (assuming aliasing did not occur).

While this method is mathematically pure, it is difficult to generate the required narrow pulses in electronics. To get around this, nearly all DACs operate by holding the last value until another sample is received. This is called a **zeroth-order hold**, the DAC equivalent of the sample-and-hold used during ADC. (A first-order hold is straight lines between the points, a second-order hold uses parabolas, etc.). The zeroth-order hold produces the staircase appearance shown in (c).

In the frequency domain, the zeroth-order hold results in the spectrum of the impulse train being *multiplied* by the dark curve shown in (d), given by the equation:

EQUATION 3-1
High frequency amplitude reduction due to the zeroth-order hold. This curve is plotted in Fig. 3-6d. The sampling frequency is represented by f_S. For $f = 0$, $H(f) = 1$.

$$H(f) = \left| \frac{\sin(\pi f/f_s)}{\pi f/f_s} \right|$$

This is of the general form: $\sin(\pi x)/(\pi x)$, called the **sinc function** or **sinc(x)**. The sinc function is very common in DSP, and will be discussed in more detail in later chapters. If you already have a background in this material, the zeroth-order hold can be understood as the convolution of the impulse train with a rectangular pulse, having a width equal to the sampling period. This results in the frequency domain being *multiplied* by the Fourier transform of the rectangular pulse, i.e., the sinc function. In Fig. (d), the light line shows the frequency spectrum of the impulse train (the "correct" spectrum), while the dark line shows the sinc. The frequency spectrum of the zeroth order hold signal is equal to the product of these two curves.

The analog filter used to convert the zeroth-order hold signal, (c), into the reconstructed signal, (f), needs to do two things: (1) remove all frequencies above one-half of the sampling rate, and (2) boost the frequencies by the reciprocal of the zeroth-order hold's effect, i.e., *1/sinc(x)*. This amounts to an amplification of about 36% at one-half of the sampling frequency. Figure (e) shows the ideal frequency response of this analog filter.

This *1/sinc(x)* frequency boost can be handled in four ways: (1) ignore it and accept the consequences, (2) design an analog filter to include the *1/sinc(x)*